

Adopted Levels, Gammas

$Q(\beta^-)=3988.4$ 19; $S(n)=8380.4$ 20; $S(p)=12155.1$ 20; $Q(\alpha)=-12332.0$ 29 2021Wa16

$S(2n)=14663.1$ 22, $S(2p)=30938$ 7 (2021Wa16).

Isotope discovery (2012Th10): $^{232}\text{Th}(^{40}\text{Ar},X)$ at Dubna (1971Ar32) and $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ at Mainz (1971Gr53).

^{35}P production:

2012Kw02: $^9\text{Be}, ^{\text{nat}}\text{Ni}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=140$ MeV/nucleon at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion- ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ and $(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

1997Vo03: $^{56}\text{Fe}(p,X)$ at $E_p=800$ MeV at LANL. Measured γ radiation. Deduced production cross sections.

^{35}P decay measurements:

1986Wa22: ^{35}P produced via $^{36}\text{S}(t,\alpha)^{35}\text{P}$ at BNL. Measured β -delayed γ rays.

1972Ap01: ^{35}P produced via $^{37}\text{Cl}(t,\alpha p)^{35}\text{P}$ at LANL. Measured $T_{1/2}$ and β -delayed γ rays.

1972Go31: ^{35}P produced via $^{18}\text{O}(^{19}\text{F},2p)^{35}\text{P}$ and $^{36}\text{S}(t,\alpha)^{35}\text{P}$ at BNL. Measured $T_{1/2}$ and β -delayed γ rays.

1971Gr53: ^{35}P produced via $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ at Mainz. Measured $T_{1/2}$ and γ rays.

^{35}P radius measurement:

1999Ai02: $\text{Si}(^{35}\text{P},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

^{35}P mass measurements:

$^{34}\text{S}(^{18}\text{O},^{17}\text{F})$ and $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$ (1988Or01), $^{36}\text{S}(^6\text{Li},^7\text{Be})$ (1985Dr06), $^{36}\text{S}(d,^3\text{He})$ (1985Kh04), $^{36}\text{S}(^{14}\text{C},^{15}\text{N})$ (1984Ma49).

Theoretical calculations (binding energies, dipole moments, quadrupole moments, radii, levels, J, π , etc.): 2012BoZT, 2009No01, 2004Kh16, 2003Sm02, 1999Du05, 1988Wa04, 1987Wa10, 1986Wo02, 1983Wi08, 1975JeZX.

 ^{35}P LevelsCross Reference (XREF) Flags

A	$^{35}\text{Si} \beta^-$ decay (0.78 s)	E	$^2\text{H}(^{36}\text{S},^3\text{He})$	I	$^{36}\text{S}(d,^3\text{He})$
B	$^{36}\text{Si} \beta^-$ decay (503 ms)	F	$^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$	J	$^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$
C	$^1\text{H}(^{34}\text{Si},p)$:resonances	G	$^{34}\text{S}(^{18}\text{O},^{17}\text{F})$	K	$^{160}\text{Gd}(^{37}\text{Cl},X\gamma)$
D	$^2\text{H}(^{34}\text{Si},^{35}\text{P}\gamma)$	H	$^{36}\text{S}(\text{pol } d,^3\text{He})$	L	$^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$

$E(\text{level})^\dagger$	J^π	$T_{1/2}$ or $\Gamma^\#$	XREF	Comments
0 1	$1/2^+$	47.3 s 8	A DEFGHIJKL	$\% \beta^- = 100$ J^π : L(pol d, ^3He)=0 from 0^+ and analyzing power. $T_{1/2}$: weighted average of 45 s 2 (1971Gr53, $\gamma(t)$), 47.7 s 29 (1972Go31, $\gamma(t)$), 48.2 s 17 (1972Go31, $\gamma(t)$), and 47.4 s 8 (1972Ap01, $\gamma(t)$).
2386.9 11	$3/2^+$	<0.69 ps	A DEFGH J L	XREF: G(2420) J^π : L(pol d, ^3He)=2 from 0^+ and L-1/2 transfer from analyzing power.
3860.4 11	$5/2^+$	<0.69 ps	A DEF HIJKL	J^π : L(pol d, ^3He)=2 from 0^+ and L+1/2 transfer from analyzing power.
4101.7 11	$(7/2^-)^{\ddagger}$	>69 ps	A D F KL	
4250 20			J	
4382.0 12	$(5/2^-)$		A F L	XREF: A(?) J^π : possibly allowed β^- feeding from $7/2^-$ parent; 1994.9 γ to $3/2^+$.
4494.1 12	$(7/2^-)^{\ddagger}$	2.29 ps 49	A D F H KL	XREF: H(4474) J^π : L=(3) in $^9\text{Be}(^{36}\text{S},^{35}\text{P})$ from 0^+ .
4666.2 16	$5/2^+$		EF HIJ	XREF: J(4640) J^π : L(pol d, ^3He)=2 from 0^+ and L+1/2 transfer from analyzing power.

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Adopted Levels, Gammas (continued)

^{35}P Levels (continued)					
E(level) [†]	J ^π	T _{1/2} or Γ [#]	XREF		Comments
4767.0 13	(9/2 ⁻) [‡]		F	L	
4869.6 12	(5/2 ⁻ , 7/2 ⁻)		A D	L	J ^π : possibly allowed β ⁻ feeding from 7/2 ⁻ parent; 1009.7γ to 5/2 ⁺ .
4962.8 12	(9/2 ⁻) [‡]		A F	L	XREF: A(?)
5010 20			J		
5090.2 13	(11/2 ⁻) [‡]		FG	L	XREF: G(5070)
5199.3 16	5/2 ⁺		EF HI J		XREF: J(5220)
					J ^π : L(pol d, ^3He)=2 from 0 ⁺ and L+1/2 transfer from analyzing power.
5487.9 13				L	
5561.0 13	(5/2 ⁻)		A	L	J ^π : possibly allowed β ⁻ feeding from 7/2 ⁻ parent; 3173.5γ to 3/2 ⁺ .
5709.5 23	(1/2 ⁻)		EF		J ^π : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 ⁺ ; interpreted as the deeply bound 1p _{1/2} proton removal from 0 ⁺ in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$; 5709γ to 1/2 ⁺ .
5.86×10 ³ 5			G J		XREF: G(5890)J(5840)
					E(level): weighted average of 5890 70 from ($^{18}\text{O}, ^{17}\text{F}$) and 5840 50 from ($^{11}\text{B}, ^{13}\text{N}$).
6222.7 13	(7/2 ⁻ , 9/2, 11/2 ⁻)			L	J ^π : 1132γ to (11/2 ⁻) and 1729γ to (7/2 ⁻).
6440 60			G		
7050 60			G		
7440 60			G		
7526.9 23	(1/2 ⁻)		F H		XREF: H(7520)
					J ^π : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 ⁺ ; interpreted as the deeply bound 1p _{1/2} proton removal from 0 ⁺ in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$; 7526γ to 1/2 ⁺ .
7590 20				J	
7920 60			G		
8390 40				J	
8.60×10 ³ 10			G		
9290 50			G		
14938 24	1/2 ⁺	<12.7 keV	C		J ^π : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15161 3	5/2 ⁻ , 7/2 ⁻	<4.4 keV	C		J ^π : L=3 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15306 24	3/2 ⁺ , 5/2 ⁺	<30.4 keV	C		J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15964 18	3/2 ⁺ , 5/2 ⁺	84 keV 25	C		J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
16145 36	1/2 ⁻ , 3/2 ⁻	0.35 MeV 9	C		J ^π : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
16605 44	1/2 ⁺	0.22 MeV 15	C		J ^π : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
17254 12	3/2 ⁺ , 5/2 ⁺	<11.6 keV	C		J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
17355 15	1/2 ⁻ , 3/2 ⁻	32 keV 22	C		J ^π : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$.

[†] From a least-squares fit to γ-ray energies for levels connected with γ transitions, from particle-transfer reactions for other levels, or from proton elastic scattering for resonances.

[‡] Comparisons with shell-model calculations (2019Gr08).

[#] T_{1/2} from the differential recoil-distance method (2019Gr08) in $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and widths from the R-matrix analysis of ($^{34}\text{Si}, \text{p}$) for resonances, unless otherwise noted.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$									
$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Mult.	δ	$\alpha^\#$	Comments
2386.9	3/2 ⁺	2386.3 6	100	0	1/2 ⁺	[M1,E2]		0.00046 5	E_γ : weighted average of 2386.4 6 from $^{35}\text{Si } \beta^-$ decay, 2386 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 2386 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. B(M1)(W.u.)>0.0023 if M1, B(E2)(W.u.)>1.6 if E2.
3860.4	5/2 ⁺	1473.5 5	15.6 14	2386.9	3/2 ⁺	[M1,E2]		8.3×10 ⁻⁵ 13	E_γ : weighted average of 1473.4 5 from $^{35}\text{Si } \beta^-$ decay, 1473 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 1474 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : weighted average of 14.1 33 from $^{35}\text{Si } \beta^-$ decay and 15.9 14 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. B(M1)(W.u.)>0.0012 if M1, B(E2)(W.u.)>2.1 if E2.
		3860.2 10	100.0 32	0	1/2 ⁺	[E2]		1.12×10 ⁻³ 2	B(E2)(W.u.)>0.12 E_γ : weighted average of 3859.5 10 from $^{35}\text{Si } \beta^-$ decay, 3860 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 3861 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : From $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. Other: 100 7 from $^{35}\text{Si } \beta^-$ decay.
4101.7	(7/2 ⁻)	241.3 5	100 [†] 7	3860.4	5/2 ⁺	[E1]		0.000665 10	B(E1)(W.u.)<4.4×10 ⁻⁴ E_γ : weighted average of 241.4 3 from $^{35}\text{Si } \beta^-$ decay, 237 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 241 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 100 4 from $^{35}\text{Si } \beta^-$ decay.
		1714.8 6	6.6 [†] 17	2386.9	3/2 ⁺	[M2]		7.93×10 ⁻⁵ 11	B(M2)(W.u.)<0.16 E_γ : weighted average of 1714.7 6 from $^{35}\text{Si } \beta^-$ decay and 1715 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 22 5 from $^{35}\text{Si } \beta^-$ decay.
		4101.4 10	54 [†] 8	0	1/2 ⁺	[E3]		0.000924 13	B(E3)(W.u.)<4.8 E_γ : weighted average of 4100.8 10 from $^{35}\text{Si } \beta^-$ decay and 4102 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 135 8 from $^{35}\text{Si } \beta^-$ decay.
4382.0	(5/2 ⁻)	1994.9 6	100	2386.9	3/2 ⁺				E_γ : weighted average of 1994.8 6 from $^{35}\text{Si } \beta^-$ decay, 1995 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 1995 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. Placement by 1988DuZS, 2008Wi09, and 2016Mu03. 1988DuZT and 1987Wa10 placed this γ as the 6096→4101 transition. 1988Or01 placed this γ as the 6488→4493 transition.
4494.1	(7/2 ⁻)	392.3 3	100 5	4101.7	(7/2 ⁻)	[M1+E2]	<0.22	0.000199 12	B(M1)(W.u.)=0.117 +42-29 E_γ : weighted average of 392.3 3 from $^{35}\text{Si } \beta^-$ decay, 391

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Mult.	$\alpha^\#$	Comments
								2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 392 <i>I</i> from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : From $^{35}\text{Si} \beta^-$ decay. Other: 100 <i>I</i> 7 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. δ : deduced by evaluators from RUL=100 for B(E2)(W.u.). B(E1)(W.u.)=2.8×10 ⁻⁴ +8-6 E_γ : weighted average of 633.7 <i>I</i> 5 from $^{35}\text{Si} \beta^-$ decay, 634 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 633 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : weighted average of 38 <i>I</i> 5 from $^{35}\text{Si} \beta^-$ decay and 27 <i>I</i> 7 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
4494.1	(7/2 ⁻)	633.6 <i>I</i> 5	34 <i>I</i> 5	3860.4	5/2 ⁺	[E1]	4.64×10 ⁻⁵ 7	
4666.2	5/2 ⁺	804 [‡] <i>I</i> 2 2279 [‡] <i>I</i> 2 4668 [‡] <i>I</i> 2		3860.4	5/2 ⁺ 2386.9 3/2 ⁺ 0 1/2 ⁺			
4767.0	(9/2 ⁻)	273 <i>I</i> 1	40.0 [†] <i>I</i> 25	4494.1	(7/2 ⁻)			E_γ : weighted average of 274 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 273 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
		664 <i>I</i> 1	100 [†] <i>I</i> 47	4101.7	(7/2 ⁻)			E_γ : weighted average of 666 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 664 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
4869.6	(5/2 ⁻ , 7/2 ⁻)	374 [†] <i>I</i> 1 487 [†] <i>I</i> 1 767.9 <i>I</i> 4	60 [†] <i>I</i> 20 <40 [†] 100 [†] <i>I</i> 20	4494.1	(7/2 ⁻) (5/2 ⁻) (7/2 ⁻)			E_γ : weighted average of 768.0 <i>I</i> 4 from $^{35}\text{Si} \beta^-$ decay and 767 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 100 <i>I</i> 8 from $^{35}\text{Si} \beta^-$ decay.
		1009.7 <i>I</i> 5	<20 [†]	3860.4	5/2 ⁺			E_γ : weighted average of 1009.9 <i>I</i> 5 from $^{35}\text{Si} \beta^-$ decay and 1009 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 152 <i>I</i> 32 from $^{35}\text{Si} \beta^-$ decay.
4962.8	(9/2 ⁻)	468.9 <i>I</i> 4	100 [†] <i>I</i> 8	4494.1	(7/2 ⁻)			E_γ : weighted average of 468.9 <i>I</i> 4 from $^{35}\text{Si} \beta^-$ decay, 469 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 468 <i>I</i> 2 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
		859 [†] <i>I</i> 3	66 [†] <i>I</i> 9	4101.7	(7/2 ⁻)			
5090.2	(11/2 ⁻)	128 <i>I</i> 1	50 [†] <i>I</i> 25	4962.8	(9/2 ⁻)			E_γ : weighted average of 127 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 128 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
		322 <i>I</i> 1	100 [†] <i>I</i> 35	4767.0	(9/2 ⁻)			E_γ : weighted average of 321 <i>I</i> 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and 322 <i>I</i> 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$.
5199.3	5/2 ⁺	1337 [‡] <i>I</i> 2 2811 [‡] <i>I</i> 2 5202 [‡] <i>I</i> 2		3860.4	5/2 ⁺ 2386.9 3/2 ⁺ 0 1/2 ⁺			
5487.9		993 [†] <i>I</i> 1 1387 [†] <i>I</i> 1	100 [†] <i>I</i> 20 60 [†] <i>I</i> 20	4494.1	(7/2 ⁻) (7/2 ⁻)			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}</u>	<u>I_{γ}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Comments</u>
5561.0	(5/2 ⁻)	1459.4 7	34 12	4101.7	(7/2 ⁻)	E _{γ} : weighted average of 1459.7 5 from ³⁵ Si β^- decay and 1458 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ P γ). I _{γ} : From ³⁵ Si β^- decay.
		3173.5 10	100 17	2386.9	3/2 ⁺	3173.5 γ is not observed in ²⁰⁸ Pb(³⁶ S, ³⁵ P γ), but the weaker 1459 γ deexciting the same level is observed in ²⁰⁸ Pb(³⁶ S, ³⁵ P γ). Further experiments are needed to resolve the discrepancy. E _{γ} ,I _{γ} : From ³⁵ Si β^- decay.
5709.5	(1/2 ⁻)	5709 [‡] 2		0	1/2 ⁺	
6222.7	(7/2 ⁻ ,9/2,11/2 ⁻)	1132 [†] 1	<25 [†]	5090.2	(11/2 ⁻)	
		1260 [†] 1	100 [†] 25	4962.8	(9/2 ⁻)	
		1729 [†] 1	100 [†] 25	4494.1	(7/2 ⁻)	
7526.9	(1/2 ⁻)	7526 2		0	1/2 ⁺	

[†] From ²⁰⁸Pb(³⁶S,³⁵P γ).

[‡] From ⁹Be(³⁶S,³⁵P γ).

Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level

